Population -> sample -> statistic -> sampling distribution

1 RECOGNISE IT

- Repeated random samples of the same size are imagined.
- The random object is a statistic such as $\bar{X}$ or p.
- Asked about how a statistic varies across samples.

2 DECISION / METHOD

Population (μ,σ,π) -> draw sample of size n -> compute statistic ($\bar{X}$,p) -> repeat conceptually -> distribution of statistic.

3 FORMULA

population variable: X
sample statistic: $\bar{X}$ or p
sampling distribution: all possible statistic values for fixed n

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and their standard errors under conditions.
Still know: the four-stage chain and which random object the probability concerns.

5 EXCEL ROUTE

Use Excel only after naming the object:
- raw sample mean: =AVERAGE(range)
- sample size: =COUNT(range)
- probability model comes from the sampling distribution.

6 TINY WORKED EXAMPLE

Population values generate many samples. If each sample has n=25, every sample gives a different $\bar{X}$; those $\bar{X}$ values form the sampling distribution.

7 TOP EXAM TRAPS

- Calling the sample histogram the sampling distribution.
- Using raw-X SD when the event is about $\bar{X}$.
- Confusing parameter μ with statistic $\bar{X}$.

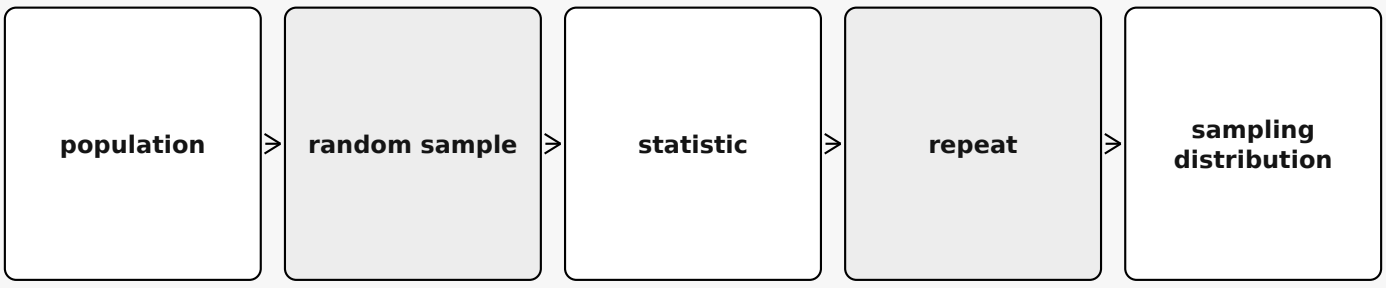
8 SAY ALOUD

A sampling distribution is the distribution of a statistic over all possible samples of one fixed size.

9 EXIT CHECK

In $P(\bar{X}>10)$, what is random: one observation or the sample mean?

ONE-GLANCE VISUAL ROUTE



Unbiasedness + exact sampling distributions

1 RECOGNISE IT

- Small finite population; asked to list all possible samples/statistics.
- Sampling rule and replacement status are specified.
- Asked whether an estimator is unbiased.

2 DECISION / METHOD

1. List all allowed samples under the stated sampling rule.
2. Compute the statistic for each sample.
3. Build its probability/frequency distribution.
4. Compare the mean of that distribution with the parameter.

3 FORMULA

`unbiased if E(estimator)=parameter`

$E(\bar{X})=\mu$

$E(p)=\pi$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table pairs $\bar{X}$ with μ and p with π.

Still know: exact enumeration, probabilities, and the expected-value definition of unbiasedness.

5 EXCEL ROUTE

For small enumeration tables:

- use rows for allowed samples
- `=AVERAGE(sample_cells)`
- average the full statistic column to check the centre.

6 TINY WORKED EXAMPLE

Population {2,4}; n=1 samples give $\bar{X}$ values 2 and 4 equally. Their mean is 3, equal to population $\mu=3$, so $\bar{X}$ is unbiased.

7 TOP EXAM TRAPS

- Unbiased does not mean every estimate equals the parameter.
- Ignoring replacement/order rules.
- Checking only one sample instead of E(estimator).

8 SAY ALOUD

Unbiased means the estimator is centred correctly over repeated samples, not that every sample is correct.

9 EXIT CHECK

What equality must hold for $\bar{X}$ to be an unbiased estimator of μ?

ONE-GLANCE VISUAL ROUTE

allowed samples → **statistic each** → **exact distribution** → **E(estimator)=parameter?**

Standard error + X versus $\bar{X}$

1 RECOGNISE IT

- Compare one observation with an average of n observations.
- Asked for the spread of sample means or effect of larger n.
- Population σ is known.

2 DECISION / METHOD

1. Name the random object: X or $\bar{X}$.
2. X uses centre μ and SD σ.
3. $\bar{X}$ uses centre μ and SE $\sigma/\sqrt{n}$.
4. Larger n narrows $\bar{X}$, not the population itself.

3 FORMULA

$SD(X)=\sigma$
$SE(\bar{X})=\sigma/\sqrt{n}$
$Var(\bar{X})=\sigma^2/n$
$E(\bar{X})=\mu$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $SE(\bar{X})=\sigma/\sqrt{n}$.
Still know: SD describes individuals; SE describes statistic-to-statistic variability.

5 EXCEL ROUTE

- `=sigma/SQRT(n)`
- raw sample: `=AVERAGE(range)`
- never use STDEV.S when population σ is explicitly supplied.

6 TINY WORKED EXAMPLE

If σ=18 and n=81, `SE(Xbar)=18/SQRT(81)=2`. Individuals still have SD 18.

7 TOP EXAM TRAPS

- Using σ instead of SE for $\bar{X}$.
- Dividing by n instead of √n.
- Saying larger n reduces population SD.

8 SAY ALOUD

X varies with σ; sample means vary with σ divided by the square root of n.

9 EXIT CHECK

If n is multiplied by 4, what happens to $SE(\bar{X})$?

ONE-GLANCE VISUAL ROUTE



Sample mean, σ known -> Z

1 RECOGNISE IT

- Probability concerns $\bar{X}$, not one X.
- Population σ is explicitly known.
- Population normal, or CLT justifies approximate normality.

2 DECISION / METHOD

1. Check the $\bar{X}$ normality condition.
2. Compute SE=σ/√n.
3. Standardise the $\bar{X}$ boundary.
4. Use left/right/between normal area.

3 FORMULA

$\bar{X} \sim N(\mu, \sigma^2/n)$

$z=(\bar{x}-\mu)/(\sigma/\sqrt{n})$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z when σ known.
Still know: the event direction and probability conversion.

5 EXCEL ROUTE

Direct left tail:
`=NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE)`
Right = 1 minus; between = upper minus lower.

6 TINY WORKED EXAMPLE

μ=50, σ=12, n=36. For P($\bar{X}$<53), SE=2 and z=1.5, so probability is about 0.9332.

7 TOP EXAM TRAPS

- Using σ instead of σ/√n.
- Using t despite known population σ.
- Applying a normal model without population normality or adequate n.

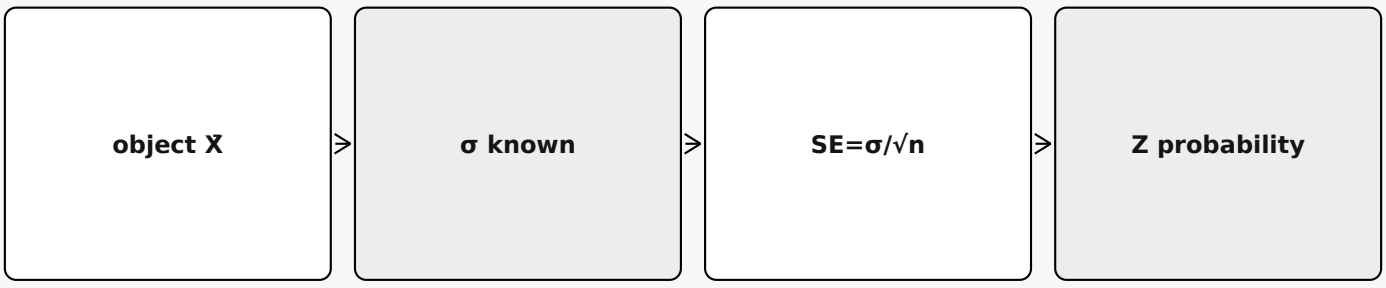
8 SAY ALOUD

Known population σ means the sample mean uses a Z model with standard error σ over square root n.

9 EXIT CHECK

What Excel formula gives P($\bar{X}$>53) for the tiny example?

ONE-GLANCE VISUAL ROUTE



Central Limit Theorem

1 RECOGNISE IT

- Population shape is not normal or is unknown.
- Probability concerns a sample mean.
- Asked whether $\bar{X}$ is approximately normal for a stated n.

2 DECISION / METHOD

1. Identify the population shape.
2. If population normal: $\bar{X}$ normal for any n.
3. Official guide: fairly symmetric often n>15; most distributions n>30.
4. Use centre μ and SE σ/√n.

3 FORMULA

for sufficiently large n:

$\bar{X}$ approx $N(\mu, \sigma^2/n)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: SE formula under conditions.
Still know: CLT concerns the sampling distribution of $\bar{X}$, not the raw population becoming normal.

5 EXCEL ROUTE

Once justified, use:
`=NORM.DIST(boundary,mu,sigma/SQRT(n),TRUE)`
with complements/differences as needed.

6 TINY WORKED EXAMPLE

A skewed population has μ=20, σ=10 and n=100. CLT gives $\bar{X}$ approximately $N(20,1^2)$.

7 TOP EXAM TRAPS

- Claiming raw observations become normal.
- Ignoring strong skew with small n.
- Forgetting that SE shrinks as n grows.

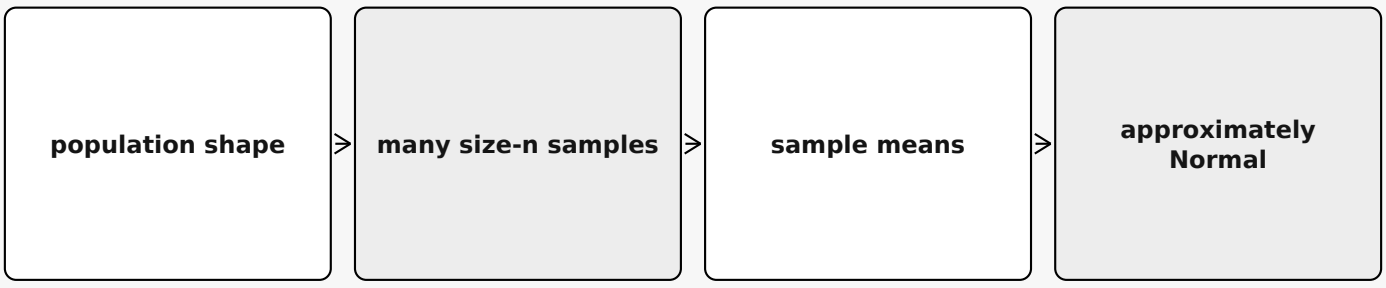
8 SAY ALOUD

The CLT makes sample means approximately normal for large n, while keeping centre μ and spread σ/√n.

9 EXIT CHECK

Does the CLT make the original population normal?

ONE-GLANCE VISUAL ROUTE



Sample mean, σ unknown -> Student t

1 RECOGNISE IT

- Probability concerns $\bar{X}$.
- Population σ is unknown and sample s is supplied.
- Population is normal/approximately normal under the course conditions.

2 DECISION / METHOD

1. Use s/√n as estimated SE.
2. Set df=n-1.
3. Convert boundaries to t.
4. Use T.DIST and the correct tail/difference.

3 FORMULA

`t=(x¯-μ)/(s/√n)`
`df=n-1`

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n, t_(n-1) when σ is unknown.
Still know: normality condition, df, and tail handling.

5 EXCEL ROUTE

- Left: `=T.DIST(tcalc,df,TRUE)`
- Right: `=1-T.DIST(tcalc,df,TRUE)`
- Between: upper T.DIST minus lower T.DIST.

6 TINY WORKED EXAMPLE

μ=30, s=5, n=25. For P(X̄<32), `t=(32-30)/(5/SQRT(25))`=2, df=24; use T.DIST(2,24,TRUE).

7 TOP EXAM TRAPS

- Using Z because n looks large while σ is unknown in this course lane.
- df=n.
- Using σ notation for sample s.

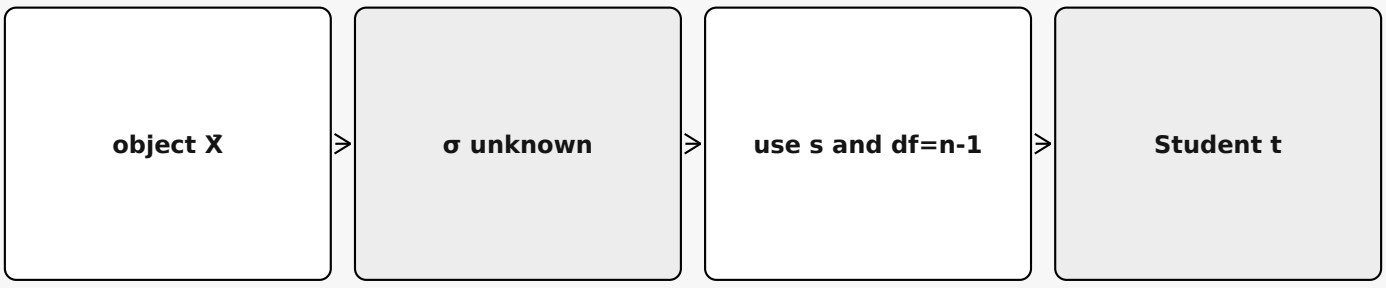
8 SAY ALOUD

Unknown σ means estimate the standard error with s and use Student t with n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function gives a cumulative-left t probability?

ONE-GLANCE VISUAL ROUTE



Sampling distribution of a proportion

A1S2 PRE-ASSESSMENT STUDY

1 RECOGNISE IT

- Each observation is success/failure.
- The random statistic is p=X/n.
- Population proportion π is known for the probability model.

2 DECISION / METHOD

1. Check nπ > 5 AND n(1-π) > 5.
2. Centre at π.
3. Use theoretical SE with π.
4. Standardise the p boundary and use normal areas.

3 FORMULA

p=X/n
E(p)=π
SE(p)=sqrt(π*(1-π)/n)
z=(p-π)/SE(p)

4 FORMULA SHEET VS WHAT I KNOW

Gives: p estimates π; theoretical SE uses π; Z model.
Still know: approximation conditions and p versus π roles.

5 EXCEL ROUTE

Direct left tail:
=NORM.DIST(p_boundary,pi,SQRT(pi*(1-pi)/n),TRUE)
Use 1-minus/right or upper-minus-lower.

6 TINY WORKED EXAMPLE

π=.40, n=100. For P(p<.50), SE=.0490 and z=2.041; probability is about .9794.

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Substituting observed p into this theoretical probability SE.
- Treating p as a count.
- Forgetting that π is fixed and p is random.

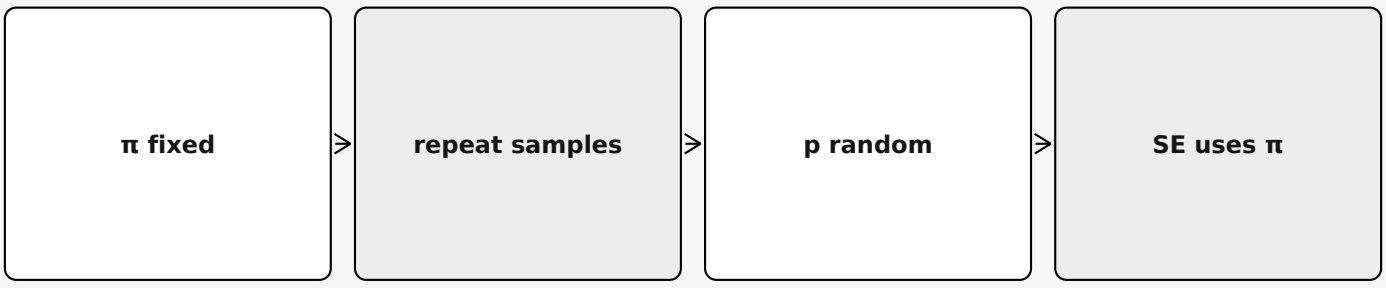
8 SAY ALOUD

For sampling probabilities, p is random, π is the centre, and theoretical SE uses π.

9 EXIT CHECK

Which symbol belongs in SE(p) for a sampling-distribution probability: p or π?

ONE-GLANCE VISUAL ROUTE



Difference between two independent sample means

1 RECOGNISE IT

- Two independent random samples are compared.
- Event concerns $\bar{X}_1-\bar{X}_2$.
- Population sigmas are known; populations normal or both n values are large.

2 DECISION / METHOD

1. Fix order: group 1 minus group 2.
2. Centre at $\mu_1-\mu_2$.
3. Combine variances, not SDs.
4. Standardise the difference and use normal areas.

3 FORMULA

$E(\bar{X}_1-\bar{X}_2)=\mu_1-\mu_2$

$SE=\sqrt{\sigma_1^2/n1+\sigma_2^2/n2}$

$z=[(d)-(\mu_1-\mu_2)]/SE$

4 FORMULA SHEET VS WHAT I KNOW

Gives: statistic, parameter difference, combined SE and Z when both sigmas are known.
Still know: independence, fixed subtraction order, and event direction.

5 EXCEL ROUTE

For $P(\bar{X}_1-\bar{X}_2>c)$:

```
=1-NORM.DIST(c,mu1-mu2,SQRT(sigma1^2/n1+sigma2^2/n2),TRUE)
```

6 TINY WORKED EXAMPLE

$\mu_1-\mu_2=5$, $\sigma_1=3$,$n1=25$,$\sigma_2=4$,$n2=16$. $SE=\sqrt{1.36}=1.166$. $P(diff>7)$ uses $z=1.715$, about .0432.

7 TOP EXAM TRAPS

- Reversing group order mid-solution.
- Adding SDs instead of variances.
- Using paired methods for independent samples.

8 SAY ALOUD

For independent means, subtract in one fixed order and add their sampling variances under the square root.

9 EXIT CHECK

What is the centre of $\bar{X}_1-\bar{X}_2$?

ONE-GLANCE VISUAL ROUTE

LOCK: group 1-group 2 → **add variances** → **divide by SE** → **Normal area**

Finite-population correction (FPC)

⚠ PROVISIONAL / LEGACY-SUPPORTED BRIDGE

1 RECOGNISE IT

- A finite population size N is supplied.
- Sampling is without replacement.
- Sampling fraction n/N is more than 0.05.

2 DECISION / METHOD

1. Confirm finite N and without replacement.
2. Compute ordinary SE.
3. Compute FPC.
4. Multiply SE by FPC before standardising.
5. If with replacement or n/N<=0.05, keep ordinary SE.

3 FORMULA

```
FPC=sqrt((N-n)/(N-1))
adjusted SE = ordinary SE * FPC
```

4 FORMULA SHEET VS WHAT I KNOW

Status: C7.52-C7.58 retain the canonical master's provisional scope safeguard. Bridge: retained as a sourced bridge, especially into Chapter 8; not silently promoted as current Chapter 7 core.

5 EXCEL ROUTE

- `=SQRT((N-n)/(N-1))`
- adjusted mean SE: `=sigma/SQRT(n)*FPC`
- adjusted p SE: theoretical SE times FPC.

6 TINY WORKED EXAMPLE

N=500,n=50 without replacement: `FPC=SQRT(450/499)=0.9496`. Multiply the ordinary SE by 0.9496.

7 TOP EXAM TRAPS

- Applying FPC merely because N is stated.
- Applying it when sampling with replacement.
- Omitting the square root.
- Failing to inspect n/N.

8 SAY ALOUD

Use FPC only for material sampling without replacement from a finite population: inspect replacement and n/N first.

9 EXIT CHECK

N=1000,n=20 without replacement: does the 5% rule trigger FPC?

ONE-GLANCE VISUAL ROUTE

